MIL-R-28819(EC)
19 May 1975
SUPERSEDING (See 6.5)

MILITARY SPECIFICATION

RADIO NAVIGATION RECEIVING SET, OMEGA, AN/BRN-7, AN/SRN-17

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Purpose. This specification prescribes the requirements for the design, construction, and testing of an OMEGA radio navigation receiving set for submarines and surface ships.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for (Navy)
MIL-E-16400	- Electronic Equipment, Naval Ship and Shore; General Specification (Navy)
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	- Definitions of and Basic Requirement for Enclosure for Electric and Electronic Equipment
MIL-STD-167	- Mechanical Vibrations of Shipboard Equipment
MIL-STD-461	- Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of
MIL-STD-480	- Configuration Control-Engineering Changes, Deviations and Waivers

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MIL-STD-781	- Reliability Tests Exponential Distribution
MIL-STD-1399	- Interface Standard for Shipboard Systems
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment, and Facilities

PUBLICATIONS

NAVAL ELECTRONIC SYSTEMS COMMAND

NORT 71-71	- Submarine OMEGA Computer Program (Performance Specification)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. Unless otherwise specified herein, the equipment shall meet the requirements of MIL-E-16400.

3.2 Parts and materials. In the selection of parts and materials, the fulfillment of design objectives shall be the prime consideration. Microelectronic technology shall be used.

3.2.1 Nonstandard parts and materials. Approval for the use of nonstandard parts and materials shall be obtained in accordance with MIL-E-16400 and as specified herein. Throw-away modules shall be considered parts, and nonstandard part approval shall be obtained for each such part.

3.2.2 Modular construction of equipment. The equipment shall be modularized in accordance with MIL-E-16400 and as specified herein. Maximum use of nonrepairable, throw-away modules shall be made. Throw-away modules shall be inherently low cost and highly reliable.

3.2.3 Mercury contamination. Components shall not have come in contact with mercury, mercury compounds, or mercury bearing instruments or devices employing only a single boundary of containment during manufacture, tests, or inspections without the specific written approval of the command or agency concerned.

3.2.4 Plastic encapsulated devices. Plastic encapsulated devices shall not be used.

3.3 Design and construction. Design and construction shall be in accordance with MIL-E-16400 and subject to the approval of the command or agency concerned.

3.3.1 Cabling and connections. Cables and connectors shall be chosen in accordance with MIL-E-16400. All interconnecting cables (between units of the

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receiving set) shall be made up on 1 end and supplied with the receiving set. Mating connectors for external wiring cables shall be supplied with the receiving set. The receiver computer and interconnecting group shall be separated by not more than 5 feet (150 cm) and they shall, in turn, be separated from the control indicator by not more than 30 feet (900 cm).

3.3.2 Interchangeability of standard modules. No adjustments shall be required when replacing plug-in modules other than normal control functions associated with turn-off and turn-on of the equipment. All plug-in modules shall be designed to preclude improper insertion into the receiving set. Modules produced under this specification shall be two-way interchangeable with the corresponding modules delivered with existing AN/SRN-17 and AN/BRN-7 equipment.

3.3.3 Configuration control. Configuration control shall be maintained in accordance with MIL-STD-480. All engineering changes, deviations, and waivers shall be subject to the approval of the command or agency concerned.

3.4 Reliability. The receiving set shall have a mean time between failure (MTBF) of at least 1000 hours, (θ_o), in accordance with MIL-STD-781.

3.5 Maintainability. The receiving set specified herein shall have an equipment repair time (ERT) of no greater than 30 minutes when repair is by replacement of modules and parts at the organizational level. The equipment repair time includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all module and part repairs.

3.6 Environmental conditions. Environmental conditions shall be in accordance with MIL-E-16400 and as specified herein. All parts of the equipment except the AS-2960/SRN-17 antenna are not exposed to weather or external air pressure.

3.6.1 Temperature classification. Temperature classification of the equipment shall be range 4 of MIL-E-16400.

3.6.2 Wind speed, icing, hydrostatic pressure, hydraulic shock, fungus, magnetic field environment, and nuclear air blast. The wind speed, icing hydrostatic pressure, hydraulic shock, fungus, magnetic field environment, and nuclear air blast requirements of MIL-E-16400 are not applicable to the specified equipment.

3.6.3 Electromagnetic interference (EMI). Electromagnetic interference requirements shall be in accordance with MIL-STD-461 and as specified herein. The equipment shall be considered class IA.

3.6.4 Shock. Shock requirements shall be in accordance with MIL-S-901. Grade A shockproofness shall be required. The equipment shall be classified as deck mounted.

3.6.5 Vibration. Vibration requirements shall be in accordance with MIL-STD-167 and as specified herein. The equipment shall meet the type I requirements up to and including 33 hertz (Hz).

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3.6.6 Enclosure. Enclosure requirements shall be in accordance with MIL-STD-108 and as specified herein. The degree of enclosure shall be dripproof at a 15° angle.

3.7 Electrical design. Electrical design shall be in accordance with MIL-E-16400 and as specified herein.

3.7.1 Power. Primary power will be provided by a single interface between the system of the ship and the interface box. All other power requirements for the equipment shall be derived from this primary power input. The interconnecting group shall contain a power ON-OFF switch and fuses (with blown fuse indicators). The maximum power shall not exceed 400 watts. The cable type for the power input will be FHOF-3. The green lead and pin B of the connector shall be equipment ground.

3.7.1.1 Primary power supply line voltage and frequency. Primary power will be 115 volts (V), 400 Hz, single phase. Regulation shall be in accordance with MIL-E-16400.

3.7.1.2 Orderly shutdown. The receiver computer power supply assembly shall provide for an orderly shutdown in the event of primary power loss. The orderly shutdown shall prevent loss of stored data in the computer and shall prevent permanent damage to the receiving set which could be caused by power transients. Upon return of primary power after a dropout, the receiver shall automatically re-initiate the synchronization routine, and then shall continue navigation (either using OMEGA signals or dead reckoning as appropriate) using the positional and time information which was available immediately prior to the power loss.

3.7.2 Interface requirements. The AN/BRN-7 shall interface with the orthogonal loop and floating wire antennas of the ship, via the CU-1441/BRR multicoupler, and the heading and speed references of the ship. The AN/SRN-17 shall interface with an AS-2960 whip antenna.

3.7.2.1 Input signals. The OMEGA signals from the submarines orthogonal loop antenna (AT-317/U) and floating wire antenna (AS-1554/U) are routed to the navigation set through the CU-1441/BRR multicoupler in the case of the AN/BRN-7 and from the AS-2960 whip antenna in the case of the AN/SRN-17. In the AN/BRN-7, switching functions are performed on the OMEGA signals internal to the CU-1441/BRR and the output signals are routed to the receiver on a single RG-264/U cable with a nominal 50 ohm impedance balanced to ground. The fore/aft loop of the orthogonal loop antenna and the floating wire antenna utilize the same two conductors of the RD-264A/U cable and are electrically oriented parallel to the keel of the ship. The athwart loop of the orthogonal loop antenna is electrically oriented perpendicular to the keel of the ship. Both antennas shall be used only in the active (on the CU-1441/BRR multicoupler) mode.

3.7.2.2 Heading and speed references of the ship. Heading of the ship and speed of the ship will be coupled to the receiver as synchro inputs. The synchro references will be 115V, 400 Hz. Synchro excitation will be 90V line to line.

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For the heading of the ship, a single revolution, 0 to 360 degrees, synchro will be used. For the speed of the ship a single revolution, 0 to 40 knots, synchro will be used for the AN/BRN-7. For the AN/SRN-17, 0 to 100 knots per revolution will be used. Synchro interface requirements shall be in accordance with Section 702 of MIL-STD-1399. Synchro inputs to the box shall be via an isolation switch which is an integral part of the box. The isolation switch shall be capable of deenergizing harmful voltages at the point of entry to the ON-128. The cable type for the synchro inputs shall be 35A-7.

3.7.3 Loop resolver. The submarine antenna systems specified herein are directional. The AN/BRN-7 shall be designed to incorporate loop resolving circuitry to correctly switch the input signals from the CU-1441/BRR multicoupler to the receiver.

3.7.4 Antenna selection indicator. A switch shall be provided on the C-9461 control box which will enable the operator to indicate to the antenna switching matrix which antenna (loop or floater) is in use. The switch shall be of the switch indicator type and it shall be mounted as an integral part of the control-indicator.

3.7.5 Timing. All timing and phase measurement functions shall be derived from a stable internal frequency standard. The standard shall meet the following requirements:

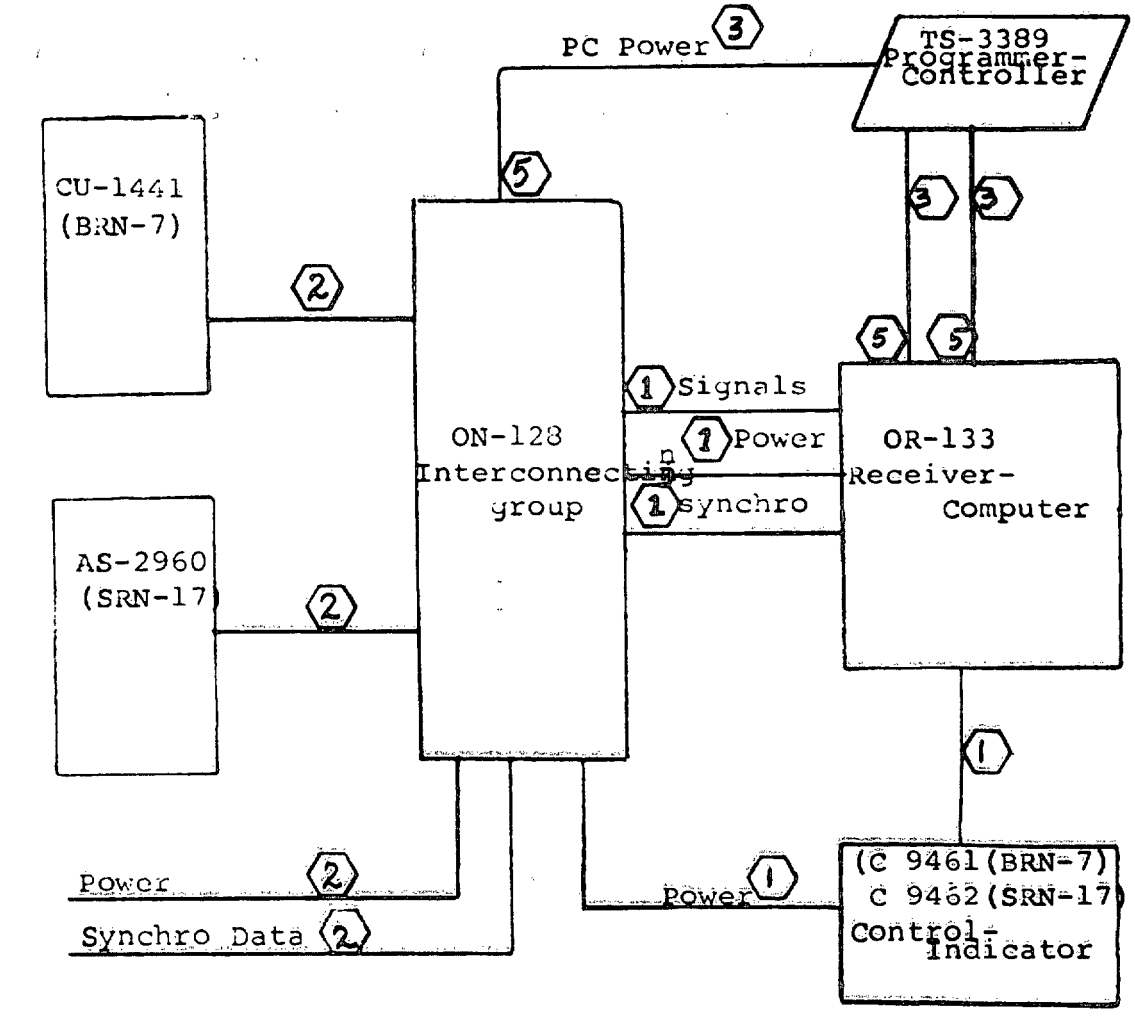
(a) Long time stability shall be, at least, 1 part in 10^8 per day.
(b) Self calibration of the standard shall be accomplished by a compensating computer program routine which derives the appropriate corrections from the off-the-air signals.

3.8 Mechanical design. The mechanical design shall be in accordance with MIL-E-16400 and as specified herein.

3.8.1 Mechanical functionalization. The receiving set shall be mechanically functionalized as shown in figure 1. Functions of the units are as follows:

(a) The ON-128 interconnecting group shall provide the necessary hardware to interface the receiver computer and control indicator with the systems of the ship. It also shall provide a power output connector to provide power to the programmer controller, TS-3389.
(b) The OR-133 receiver computer shall contain the hardware to detect and measure the OMEGA signals and make the necessary computations to meet the functional performance requirements specified herein.
(c) The C-9461 and C-9462 control indicators shall provide the man-machine interface between the operator and the receiving sets. Functions of the controls and indicators shall be as described in 3.9.
(d) (AN/SRN-17 only) The AS-2960 antenna shall receive transmitted OMEGA signals.

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Notes:
1. Cable supplied
2. Cable not supplied
3. Cable part of TS-3389
4. Mating Connector supplied with CU-1441
5. Mating connector supplied with TS-3389

Figure 1. OMEGA navigation receiving sets, AN/BRN-7 and AN/SRN-17.

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3.8.2 Size and weight. Maximum weight of the individual units are as Table I (all weights and dimensions include mounting bases).

Table I. Size and weight.

Unit	Height (in)	Height (mm)	Width (in)	Width (mm)	Depth (in)	Depth (mm)	Weight (lbs)
Receiver computer	11.5	(292.10)	13.1	(332.74)	23.2	(589.28)	68
Control indicator	13.2	(335.28)	9.2	(233.68)	8.6	(216.44)	17
Interface box	9.2	(233.68)	9.0	(228.60)	11.0	(279.40)	36
Antenna	125.3	(3182.62)	5.5	(139.70)	6.0	(152.40)	9

3.8.3 Mounting bases. Access to replaceable or repairable assemblies, in the control indicator and receiver computer units, shall be provided by the use of suitable mounting bases to be supplied as part of the receiving set.

3.8.4 Test equipment access. Provision shall be made for interface of the receiving set with the programmer controller. A maximum of three cable type interface connections may be used. Quick disconnect connectors shall be provided at convenient locations on the receiver computer front panel for this function.

3.9 Controls, indicators, and panel layout. Controls, indicators, and panel layout shall be in accordance with MIL-E-16400 and as specified herein.

3.9.1 Panel layout. The front panel layout of the control indicator shall be as shown in Figure 2. Descriptions and characteristics of the controls and indicators shown in Figure 2 are given in 3.9.

3.9.2 Indicators. Lamps and other indicators used on the front panels shall be in accordance with MIL-STD-1472 and as specified herein. Indicators shall have opaque backgrounds with illuminated red legends except the SYSTEM MALFUNCTION indicator which shall have a luminous red background with opaque legend.

3.9.3 Edge lighting. Edge lighting of the control indicator is not required.

3.10 Performance characteristics. The overall performance of the receiving set shall be as specified herein. No deviation in performance shall be made to the equipment without the written approval of the command or agency concerned.

3.10.1 Functional characteristics. As specified herein, the receiving set shall have functional characteristics categorized as either overall or specific.

3.10.1.1 Overall functional characteristics. The receiving set specified herein shall have as its primary functional capability the following:

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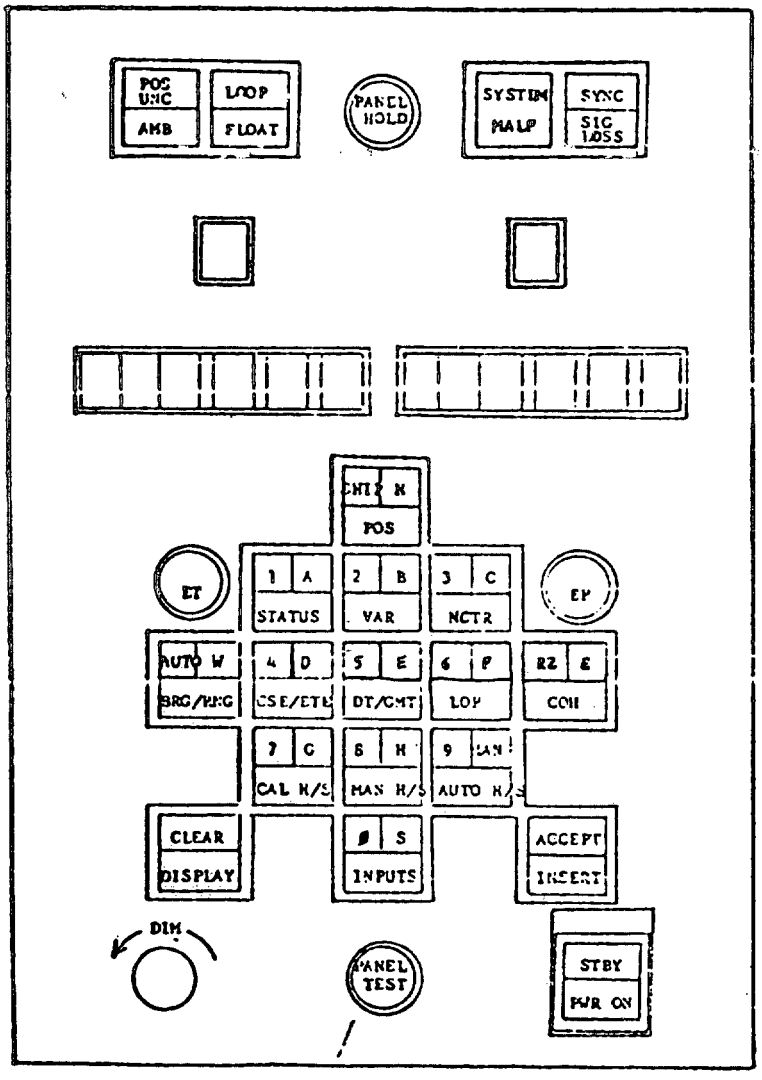


Figure 2. Control indicator panel layout C-9461 (AN/BRN-7) C-9462 (on AN/SRN-17 the loop/float switch is blank).

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(a) Station/frequency usage. The set shall have the capability of using all available minimal path signals from the eight transmitting stations, when available, on the three primary navigation frequencies of 10.2 kilohertz (kHz), 11 1/3 kHz, and 13.6 kHz.

(b) Automatic synchronization. The set shall have the capability of synchronizing with the OMEGA signal format for purposes of recognizing and using the transmissions from the various stations. Synchronization shall be accomplished without operator assistance, other than normal turn-on, and synchronization shall be maintained during periods of time when OMEGA signals are not available.

(c) World-wide capability. The set shall provide continuous navigation anywhere on the surface of the earth when all eight transmitting stations are in operation.

(d) Tracking. The set shall track up to eight stations using all three navigation frequencies without operator assistance.

(e) Phase measurement. The set shall determine position by making phase measurements on the OMEGA transmissions.

(f) Dead reckoning. The set shall automatically dead reckon when OMEGA signals cannot be received.

3.10.1.2 Specific functional characteristics. Performance requirements for specific functions shall be as specified in (a) through (g).

(a) Latitude and longitude. Latitude and longitude shall be displayed in degrees and minutes with a resolution to 0.1 minute. The accuracy of latitude and longitude calculations, not including that for the position of the ship, shall be ±0.1 minute.

(b) Velocity. Velocity shall be displayed in knots with a resolution to 0.1 knot. The accuracy of velocity calculations, not including the calculated velocity and heading display, shall be ±0.1 knot.

(c) Heading. Heading shall be displayed in degrees with a resolution to 1 degree. The accuracy of heading calculations, not including the calculated velocity and heading display, shall be ±1 degree. All headings displayed and inserted shall be true heading.

(d) Distance. Distance shall be displayed in nautical miles with a resolution to 0.1 nautical mile. The accuracy of the distance calculations shall be ±0.1 nautical mile.

(e) Estimated time enroute. Estimated time enroute shall be displayed in hours with a resolution to 0.1 hour. The accuracy of the estimated time enroute calculations shall be ±0.1 hour.

(f) Date and greenwich mean time. Date and Greenwich mean time shall be displayed in year, month, date, hour, minutes and seconds with a resolution to one second. The accuracy of the time calculation shall be ±1 second.

(g) Line of position. Line of position (see 6.2.3) shall be displayed in lane number and percent of a lane with a resolution to one percent of a lane; one centicycle (cec) (see 6.2.4).

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3.10.2 Signal acquisition. The receiving set shall be capable of acquiring the OMEGA signals, automatically synchronizing its internal timing circuits with the signals, and tracking the signals under the following signal conditions:

(a) At least three station signals are receivable on all three frequencies.
(b) The signal to noise (S/N) ratio is at least 0 decibels (dB) for one station's signals and at least -15 dB for the other two stations' signals.

3.10.3 Dynamic range. The specified dynamic range shall be 5 microvolts (mV) per meter to 500,000 mV per meter. With one channel of the receiver held at a constant input signal level, performance of the receiving set, measured at the output of the burst measurement filters, shall be as in Table II.

Table II. Dynamic range.

Variable input signal (V/m)	Mean difference from absolute (cec)	Standard deviation (cec)
5 uV/M	±3	5
50 uV/m	±2	3
500 uV/m	±1	1
5 mV/m	±1	1
50 mV/m	±2	3
500 mV/m	±3	5

3.10.4 Random noise. The receiving set shall be capable of using signals in the presence of random noise. Random noise shall be measured in a bandwidth of 100 Hz. Performance of the receiving set, measured at the output of the burst measurement filters, shall be as in Table III.

Table III. Random noise.

S/N ratio (dB)	Mean difference from absolute (cec)	Standard deviation (cec)
+20	±1	1.5
+10	±1	1.5
0	±1	2
-10	±2	3
-15	±3	5

3.10.5 Navigation performance. The primary function of the receiver shall be navigation using available OMEGA signals. The receiver shall calculate the appropriate lines of position, apply appropriate propagation predictions and coordinate conversions, and compute the absolute position of the receiver. The optimum position determination calculated by the receiving set shall have an accuracy of 1 to 2 nautical miles root mean square (rms).

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3.10.6 Input functions. The receiving set shall be capable of using the following information input to the computer through the control indicator panel:

(a) Geodetic latitude and longitude coordinates (POS) of the ship (SHIP) with a weighting factor of the accuracy of the position; better than 0.5 mile (A), better than 5 miles (B), better than 35 miles (C).
(b) Geodetic latitude and longitude coordinates (POS) of 10 destination points (1, 2, 3, 4, 5, 6, 7, 8, 9, 0).
(c) Geodetic latitude and longitude coordinates (POS) of a rendezvous (RZ).
(d) Velocity and heading (MAN V/H) of the ship (SHIP).
(e) Velocity and heading (MAN V/H) of the rendezvous (RZ).
(f) Calendar date and Greenwich mean time (DT/GMT).
(g) Deselection (INPUTS) of OMEGA signals from any given station (A, B, C, D, E, F, G, H) and the unused velocity and heading mode (MAN, AUTO).
(h) Selection of the antenna switching procedure for floating wire (FLOAT) or orthogonal loop (LOOP) to be used by the antenna switching matrix (for the AN/BRN-7 only).

3.10.7 Display functions. The receiving set shall be capable of displaying the following functions through the control indicator panel:

(a) Geodetic latitude and longitude (POS) of the ship, any 1 of 10 destination points and the rendezvous point (RZ).
(b) S/N quality (STATUS) of the OMEGA signals for 10.2 kHz (1), 11 1/3 kHz (2), 13.6 kHz (3) or predicted coherency (COH) of the OMEGA signals between any of the eight stations (A, B, C, D, E, F, G, H) and the base station.
(c) Position variance (VARIANCE) of the Kalman filter.
(d) Kalman filter update rate (NCTR) for each station (A, B, C, D, E, F, G, H).
(e) Range and bearing (RNG/BRG) between the ship, one of the ten destination points or between the rendezvous point (RZ) and one of the ten destination points or between the ship and the rendezvous point (RZ).
(f) Estimated time enroute at present speed and course (ETE/CSE) between the ship or one of the ten destination points and one of the ten destination points or the rendezvous point (RZ).
(g) Calendar date and Greenwich mean time (DT/GMT).
(h) Uncorrected OMEGA line of position (LOP) between two pairs of OMEGA stations on 10.2 kHz, 11 1/3 kHz and 13.6 kHz.
(i) Coherency (COH) of the OMEGA signals between any of the eight stations and the base station.
(j) Calculated speed and heading (CAL/H) of the ship.
(k) Manual speed and heading (MAN/H) of the ship or the rendezvous (RZ).
(l) Automatically inserted speed and heading (AUTO V/H) of the ship derived from the heading of the ship and speed reference inputs.
(m) OMEGA stations, base station, and heading and speed references being used by the receiver (INPUTS).

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3.10.8 Warning indicators, status indicators, test functions, and controls. The receiving set shall be capable of displaying or controlling operation through the following indicators and controls of the control indicator panel.

(a) Display position uncertainty (POS UNC) warning when the receiver is in the difference frequency mode.
(b) Display ambiguity (AMB) warning when more than one state vector exists in the Kalman filter.
(c) Display the antenna, floating wire (FLOAT), or orthogonal loop (LOOP) being used by the AN/BRN-7.
(d) Display a system malfunction (SYS MALF) warning when the built-in-test-equipment detects an out of tolerance condition.
(e) Display a synchronization (SYNC) warning when the receiver is in the synchronizing mode.
(f) Display a signal loss (SIG LOSS) warning when signals are not available at the radio frequency (RF) input.
(g) Control a front panel indicator test (TEST).
(h) Control intensity (DIM) of all indicators.
(i) Stop update (HOLD) of the panel for some displays (POS, RNG/BRG, ETE/CSE, DT/GMT, CAL/H, AUTO/H).
(j) Control the standby (STBY) and operating (ON) modes.
(k) Display enter time (ET) and enter position (EP) warnings when power has been lost and it has caused an interruption to normal program flow.
(l) Control the display (DISPLAY) and clear (CLEAR) functions.
(m) Control the insert (INSERT) and accept (ACCEPT) functions.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will include (a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

(1) Production inspection (see 4.4.1)
(2) Production control inspection (see 4.4.2)
(3) Environmental inspection (see 4.4.3)
(4) Inspection of preparation for delivery (see 4.4.5)

(c) Reliability testing (see 4.5)

4.3 First article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the examination and testing specified in production inspection, production control inspection, and environmental inspection.

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process, such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall include the requirements of Table IV.

Table IV. Production inspection.

Inspection	Reference
Loop resolver	3.7.3
Antenna selection indicator	3.7.4
Timing	3.7.5
Functional characteristics (specific)	3.10.1.2
Signal acquisition	3.10.2
Dynamic range	3.10.3
Random noise	3.10.4
Navigation performance	3.10.5
Input functions	3.10.6
Display functions	3.10.7
Warning indicators, status indicators, test functions and controls	3.10.8

4.4.2 Production control inspection. Production control inspection, including sampling, shall conform to Table V and to the inspection procedures of MIL-STD-105 using the special inspection levels. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection

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shall be performed on equipments that have passed production inspection. The equipment shall satisfactorily meet the requirements of production control inspection prior to release for shipment.

Table V. Production control inspection.

Inspection	Reference
Supply line voltage and frequency	3.7.1.1
Orderly shutdown	3.7.1.2
Interface requirements	3.7.2
Size and weight (see Note 1)	3.8.2
Interchangeability	3.3.2
Heat test	MIL-E-16400
Power test	3.7.1
Enclosure (see Note 2)	3.6.6

Note 1 This test shall be performed only on samples from the first lot.

Note 2 When performed as a first article inspection and the first production control inspection lot, water shall be used. Thereafter, a visual examination may be used.

4.4.2.1 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.2.2 Reinspection of conforming production control test units. Unless otherwise specified, sample units which have been subjected to, and passed, production control inspections may be accepted on the contract provided they are resubjected to, and pass, production inspection after repair of all visible damage.

4.4.3 Environmental inspection. Environmental inspection shall be as listed in Table VI and shall be performed on units which have been subjected to and passed production inspection.

Table VI. Environmental inspection.

Inspection	Reference
Temperature	3.6
Humidity	3.6
Electromagnetic interference	3.6.3
Shock	3.6.4
Vibration	3.6.5
Inclination	3.6

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Table VI. Environmental inspection (continued).

Inspection	Reference
Structural integrity	3.1.1
Salt fog	3.6
Magnetic test	3.1.1
Enclosure	3.6.6

4.4.3.1 Sampling for environmental inspection. One equipment shall be selected from each successive 100 equipments produced or fraction thereof. The first sample equipment shall be selected from the first month's production.

4.4.3.2 Nonconforming environmental sample units. If a sample unit fails the environmental inspection, the contractor shall immediately investigate the cause of failure and report to the Naval Electronic Systems Command and the Quality Assurance Representative the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes. If the command or agency concerned or the Quality Assurance Representative does not consider the corrective action sufficient to enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the delivery of equipment to the Government may be withheld.

4.4.3.3 Reinspection of conforming environmental sample units. Unless otherwise specified, sample units which have been subjected to, and passed, environmental tests may be accepted on the contract provided they are resubjected to, and pass, production inspection after repair of all damage.

4.4.4 Inspection of preparation for delivery. Inspection shall be conducted to ensure conformance with the requirements of section 5 herein.

4.5 Reliability testing. Reliability testing of the equipment specified herein shall be performed in accordance with the requirements of MIL-STD-781.

4.5.1 Failure. All failures will be considered relevant unless determined by the command or agency concerned to be non-relevant, or caused by a condition external to the equipment under test (see 6.3).

4.5.2 Reliability production acceptance (sampling) phase. Reliability sampling testing shall be conducted throughout production to ensure production of acceptable equipment. Testing shall be in accordance with test plan XXIX of MIL-STD-781. Testing shall be at test level B with the following modifications:

(a) Duration of testing shall be 50 hours per equipment
(b) The vibration frequency shall be between 20 Hz and 33 Hz
(c) There shall be no ON/OFF cycling.

If the reliability production acceptance phase results in a reject situation, corrective action shall be taken in accordance with MIL-STD-781.

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4.6 Special test procedures. The following special test procedure requirements apply to the production, production control, and environmental inspections. The inspection category to which the special test procedure requirement applies is listed in the inspection Tables IV through VI as a special test reference.

4.6.1 General test requirement. The test procedure shall be in accordance with the applicable requirements of MIL-E-16400.

4.6.2 Special test equipment. It will be the responsibility of the contractor to supply all test equipment required for the performance of inspections specified herein.

4.6.2.1 OMEGA signal simulator. An OMEGA signal simulator will be required to perform some tests specified herein. The absolute accuracy of the simulator shall be on an order of magnitude better than the specified requirement for which it is being used.

4.6.2.2 Programmer controller. A programmer controller (TS-3389) will be required to perform some tests specified herein.

4.6.3 Timing test. The stability of the oscillator shall be measured. The test shall demonstrate the capability for self calibration of the standard using simulated OMEGA signals.

4.6.4 Navigation performance test. The navigation performance shall be demonstrated by using a field test over a period of 24 hours. Position fix samples shall be taken at least once per 30 minutes during the test and a total of at least 35 samples must be collected.

4.6.5 Electromagnetic interference. The requirements of MIL-STD-461 shall be verified by tests performed in accordance with MIL-STD-462. Susceptibility test procedures may be modified to be compatible with the performance requirements of the equipment. All modifications shall be approved by the command or agency concerned.

4.6.6 Shock test. The following clarifications of the shock test requirements are applicable:

(a) Class II testing shall be required.
(b) Test classification shall be lightweight.
(c) Type A tests shall be required.
(d) Subsequent to testing, no permanent deformation shall be visible, no permanent functional impairment shall occur, and the equipment shall meet the performance requirements specified herein. Momentary impairment of operation during the shock is acceptable.
(e) Mounting shall be in accordance with approved installation drawings.
(f) The equipment shall be energized during the test.

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4.6.7 Vibration test. The following clarifications of the vibration test requirements are applicable:

(a) Subsequent to testing, the equipment shall meet the performance requirements specified herein. During the test, the equipment shall display latitude and longitude without error in excess of the performance requirement specified herein.
(b) Mounting shall be in accordance with approved installation drawings.

4.6.8 Dynamic range and random noise test. Test data shall consist of at least 20 sample points taken in a continuous running of not less than 10 minutes for each input condition.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by the suppliers shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified, preparation for delivery shall be in accordance with MIL-E-17555.

6. NOTES

6.1 Intended use. The OMEGA navigation receiving sets specified herein will make phase measurements on the 10.2 kHz, 11 1/3 kHz, and 13.6 kHz transmissions from the OMEGA transmitting sites, process the phase measurements, and display the calculated position of the submarine or surface ship.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Marking, levels of preservation, and packaging and packing required (see 5.1).

6.3 Definitions.

6.3.1 OMEGA. OMEGA is a radio locating system providing world-wide position fixing of moderate accuracy (1 to 2 nautical miles, rms) by the phase comparison of very low frequency, continuous wave, pulsed radio signals from three or more of the eight strategically located transmitting stations.

6.3.2 OMEGA signal format. Each of the eight OMEGA stations transmits 3 navigation frequencies and a pair of timing frequencies. The entire format is repeated every 10 seconds and all transmissions are synchronized in time and phase from a central location.

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6.3.3 Line of position. Phase difference measurements between pairs of OMEGA stations' transmissions form lines of position. The zero phase difference isophase contours for a given pair of stations form a family of hyperbolae on the surface of the earth. Lanes are formed with widths of 0.5 wavelength of the carrier frequency (8 nautical miles for 10.2 kHz) on the baseline. The lanes diverge somewhat on either side of the baseline. A line of position includes the lane number and the percent of a lane.

6.3.4 Centicycle. The centicycle is used in the OMEGA system to indicate 1 percent of a lane for any 1 of the 3 carrier frequencies. The centilane (cel) represents 1 percent of a lane for the 10.2 kHz primary navigation carrier only or approximately 500 feet.

6.3.5 Programmer controller. The TS-3389 programmer controller is a special test equipment. The programmer controller interfaces with the receiver computer and performs the functions of fault isolation and computer program loading.

6.3.6 Failure. Any malfunction or deviation which prevents the equipment from performing in accordance with the performance requirements specified herein.

6.4 Maintenance concept. Maintenance of the OMEGA sets is performed at organizational and depot levels. Organizational maintenance is limited to replacement of modular assemblies and components by shipboard technicians. The depot isolates failures to the component level and performs corrective maintenance.

6.5 Supersession data. This specification includes the requirements of Naval Electronic Systems Command contract specifications ELEX-R-122, ELEX-R-125, and ELEX-R-227.

Preparing activity:
NAVY - EC

(Project No. 5825-N008)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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